PQC DAILY STUDY SERIES

PQC Adoption in the Wild: A Look at Real-World Data

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PQC Adoption in the Wild: A Look at Real-World Data

While the standardization of post-quantum algorithms like ML-KEM and ML-DSA is a monumental step, the real measure of success is their adoption in live internet protocols. A recent study from researchers at the University of Illinois at Urbana-Champaign provides a rare, data-driven look into the actual deployment of PQC in critical services like SSH, offering a crucial reality check on the global migration progress.

The PQC Network Instrument: Measuring the Transition

To understand real-world adoption, researchers developed a "PQC network instrument" to passively monitor traffic at a major supercomputing center. This allows for large-scale measurement without disrupting services, providing a unique vantage point on which cryptographic algorithms are being negotiated in the wild. The primary focus was on SSH and TLS, two of the most fundamental protocols for secure communication on the internet.

> **Key Finding:** The study revealed a nascent but growing adoption of PQC in SSH. The hybrid algorithm `sntrup761x25519-sha512@openssh.com` (a combination of a PQC algorithm and a classic Elliptic Curve algorithm) was detected, showing that major providers are beginning to experiment with and deploy quantum-resistant key exchange mechanisms. However, the overall adoption rate remains low, highlighting the significant work ahead.

Adoption Rates: A Snapshot of Progress

The research provides concrete, albeit early, metrics on the state of the PQC transition. This data is critical for understanding the gap between standards availability and real-world implementation.

Protocol	PQC Implementation Status	Observed Adoption Rate (at NCSA)	Key Takeaway
SSH (Secure Shell)	Hybrid KEX available (e.g., in OpenSSH)	~0.029% of connections	Adoption is measurable but still in its infancy. The use

Protocol	PQC Implementation Status	Observed Adoption Rate (at NCSA)	Key Takeaway
			of hybrid modes is the clear migration path.
TLS (Transport Layer Security)	Implementations exist (e.g., in Chrome via hybrid KEMs)	Not explicitly measured in this study, but noted as an area of active deployment by major browsers.	Browser-side adoption is leading, but server-side readiness remains a major hurdle.
RDP (Remote Desktop Protocol)	No simple PQC solution available	0%	Highlights a critical gap, as legacy and proprietary protocols often lag in cryptographic agility.

Challenges and Migration Pathways

The study underscores that the transition is not just about having standardized algorithms; it involves a complex ecosystem of software libraries, protocol implementations, and system administration practices. Key challenges include:

- **Server-Side Lag:** While client-side applications like browsers are adopting PQC, many servers are not yet configured to negotiate PQC-enabled connections.
- **Legacy Systems:** The continued use of deprecated classical algorithms in some connections poses an immediate security risk and indicates a lack of crypto-agility that will also hinder PQC migration.
- **Protocol Complexity:** Integrating new, larger keys and different cryptographic primitives into established protocols without breaking compatibility is a significant engineering effort.

The clear pathway forward, as demonstrated by the observed SSH traffic, is the use of **hybrid mechanisms**. This approach combines a classical and a post-quantum algorithm, ensuring security against both current and future threats and providing a safety net in case of unforeseen weaknesses in the new PQC algorithms.

Daily Quiz

1. What is the primary advantage of using a "hybrid" key exchange mechanism for PQC adoption?
2. According to the study, which internet protocol showed a measurable (though small) adoption of a PQC algorithm?
3. What is meant by "server-side lag" in the context of the PQC transition?